

01 - Server implementation

How the API becomes the source of truth for plans, trials, and entitlements.

Base: server/ Related specs: docs/02-PLAN-ENTITLEMENTS-BLUEPRINT.md,
docs/04-SUBSCRIPTION-DATA-MODEL.md

1. Current state

Piece	Status
SubscriptionPlan (country pricing, benefits, soft limits)	Done
OrganizationSubscription (Trial/Active/?)	Done
Superadmin plan CRUD + assign	Done
Structured entitlements[{key,enabled,value}]	Todo
resolveEntitlements(orgId) service	Todo
GET /api/billing/me	Todo
GET /api/public/plans	Todo
requireEntitlement / requireQuota middleware	Todo
Trial expiry cron	Todo

2. Schema changes (P0)

2.1 Extend SubscriptionPlan

Keep benefits for marketing. Add machine entitlements:

```
entitlements: [{
  key: { type: String, required: true, trim: true }, // 'bookings.crud'
  enabled: { type: Boolean, default: true },
  value: { type: mongoose.Schema.Types.Mixed, default: null }
}],

seatTypes: [{
  role: { type: String, enum: ['admin', 'manager', 'operator', 'viewer'] },
  included: { type: Number, default: 0 },
  priceExtra: { type: Number, default: 0 }
}],

trialEntitlements: [{ type: String }], // empty = same as paid
```

Migration: For existing plans, seed entitlements from category defaults (Starter / Professional / ? matrix in blueprint). Keep writing legacy features: string[] from benefit titles until Landing no longer needs it.

2.2 Optional new collections (P4-P5)

- OrgEntitlementOverride - per-org enable/disable/limit overrides
- PlanAddOn / OrgAddOn - modular extras

Ship P0-P2 without these if needed; leave hooks in the resolve service.

3. Entitlement catalog constant

Create server/src/constants/entitlementKeys.js listing every key from docs/01-ADMIN-FEATURES.md ?9.

Use it to:

- Validate Superadmin upsert payloads
- Seed default matrices by category
- Document API 403 codes

4. Resolve service

New file: server/src/services/entitlements.js

```
export async function resolveEntitlements(organizationId) {
  // 1. Load latest OrgSubscription (populate plan)
  // 2. If missing / Cancelled -> readOnlyMinimal
  // 3. If Expired -> readOnlyMinimal (+ allow billing page)
  // 4. Base = plan.entitlements (or trialEntitlements if Trial)
  // 5. Merge add-ons + overrides (when exist)
  // 6. Compute usage (count vehicles/drivers/users for org agencies)
  // 7. Return { subscription, plan, features, limits, usage, benefits }
}

export function hasFeature(resolved, key) {
  return resolved.features?.[key] === true;
}

export function checkQuota(resolved, resource) {
  // resource: 'vehicles' | 'drivers' | 'users' | ...
  // return { ok, limit, usage, code }
}
```

Usage counting

Resolve org -> primary agency / all agencies linked via organizationId (or user agencies). Count:

Resource	Count source
vehicles	Vehicle docs for agency(ies)
drivers	Driver docs
users	Users with membership on those agencies
agencies	Agency count for org

5. HTTP APIs

5.1 Authenticated billing (Admin)

Route file: server/src/routes/billing.js Mount: app.use('/api/billing', protect, billingRoutes)

Method	Path	Behavior
GET	/me	resolveEntitlements for caller's org
POST	/upgrade-request	Create PlatformLead or notify Superadmin

Org resolution for fleet user: user -> agencies -> organizationId (add organizationId on Agency if missing; or map via Project). Document the chosen link and stick to it.

5.2 Public catalog (Landing)

Route file: server/src/routes/publicPlans.js Mount: app.use('/api/public', publicPlansRoutes) - no auth

Method	Path	Behavior
--------	------	----------

GET	/plans?country=IN	Active plans for country, sorted; strip internal fields
GET	/plans/featured?country=IN	Optional featured-only

Response shape (example):

```
{
  "success": true,
  "country": "IN",
  "currency": "INR",
  "plans": [
    {
      "code": "PROFESSIONAL",
      "name": "Professional",
      "description": "...",
      "priceMonthly": 2999,
      "priceYearly": 29990,
      "pricingLabel": "",
      "trialDays": 14,
      "featured": true,
      "benefits": [{ "title": "...", "detail": "" }],
      "limits": { "vehicles": 100, "drivers": 200, "users": 15 }
    }
  ]
}
```

Rate-limit this route.

5.3 Extend Superadmin upsert

In buildPlanPayload (platformOpsController.js):

- Accept entitlements, seatTypes, trialEntitlements
- Validate keys against entitlementKeys
- Do not overwrite entitlements from benefit titles

Update seedDefaultPlans to write full entitlement matrices per tier.

6. Middleware (P2)

Files:

- server/src/middleware/requireEntitlement.js
- server/src/middleware/requireQuota.js

```
// requireEntitlement('departments.monthly_billing')
export const requireEntitlement = (key) => asyncHandler(async (req, res, next) => {
  const resolved = await resolveEntitlements(req.organizationId);
  if (!hasFeature(resolved, key)) {
    return res.status(403).json({
      success: false,
      error: 'Your plan does not include this feature.',
      code: 'FEATURE_LOCKED',
      feature: key,
      upgradeRequired: true
    });
  }
  req.entitlements = resolved;
  next();
});
```

```
// requireQuota('vehicles') before create handlers
export const requireQuota = (resource) => asyncHandler(async (req, res, next) => {
  const resolved = req.entitlements || await resolveEntitlements(req.organizationId);
  const q = checkQuota(resolved, resource);
  if (!q.ok) {
    return res.status(403).json({
      success: false,
      error: Plan limit reached for ${resource}.,
      code: QUOTA_${resource.toUpperCase()},
      limit: q.limit,
      usage: q.usage,
      upgradeRequired: true
    });
  }
  next();
});
```

Where to attach (priority)

Route area	Feature key	Quota
Vehicles create	vehicles.crud	vehicles
Drivers create	drivers.crud	drivers
User invite	-	users
Department contracts / bills	departments.*	-
Bookings create	bookings.crud	bookings (monthly, later)
Payroll	drivers.payroll	-
Multi agency create	agency.create	agencies

Attach gradually: start with vehicles + drivers + department billing.

7. Trial lifecycle (P3)

File: server/src/jobs/subscriptionLifecycle.js

Job	Schedule	Action
Expire trials	Daily	status: Trial and trialEndsAt < now -> Expired + platform event
Reminders	Daily	D-7 / D-3 / D-1 -> create Notification for primary admin

Wire from server.js with node-cron or a simple setInterval if cron not present.

On onboard (superadminController), keep auto-Trial; ensure trialEndsAt uses plan trialDays.

8. Caching

- Cache resolveEntitlements in-memory / Redis for 30-60s per org
- Invalidate on Superadmin assign / plan update / override change
- Landing public plans: cache 5 minutes per country

9. Test plan (server)

- Seed plans with entitlements for IN
- Assign Trial STARTER to org
- GET /api/billing/me as fleet admin -> features + daysLeft

- Toggle off bookings.crud -> booking create 403 FEATURE_LOCKED
 - Set limits.vehicles = 1, create second vehicle -> QUOTA_VEHICLES
 - GET /api/public/plans?country=IN -> no auth, active only
 - Expire trial manually -> resolve returns read-only
-

10. Rollout safety

- Default: if plan has empty entitlements array, treat as full access (legacy orgs) until seed runs
 - After seed: enforce
 - Feature flag env: ENTITLEMENTS_ENFORCE=true to flip hard enforcement in production
-

Next: 02-ADMIN-DASHBOARD-CONSUMPTION.md